

Experiment 9: Real-Time Bitcoin Price Analysis using a Sliding Window

Problem Statement

A financial analyst continuously monitors the live Bitcoin price using the CoinGecko API, which provides real-time cryptocurrency market data. Write a Python program to fetch the latest price at regular intervals, store the most recent values in a fixed size sliding window, and perform statistical analysis for each window to analyse short-time market fluctuations.

Aim

To fetch live Bitcoin prices from the CoinGecko API at regular intervals and perform statistical analysis on a fixed size sliding window of the most recent prices.

Theory

Cryptocurrency prices change continuously, so the price feed behaves like a data stream. Storing every price ever received is not practical, therefore only the most recent values are kept. A deque with a maxlen is used for this purpose. When the deque is full and a new price is appended, the oldest price is removed automatically, which gives a sliding window without any extra code. Statistics such as mean, median, minimum, maximum and standard deviation are then calculated on this window to understand the short-time behaviour of the market. The API call is wrapped in a try-except block, because a network request can fail or the API can return an unexpected response, and the program must continue running in either case.

Requirements

- Python 3.x
- Jupyter Notebook
- Requests
- Time module
- Collections module
- Statistics module
- An active internet connection

Step 1: Import Required Libraries

```
import requests
import time
import statistics
from collections import deque
```

Interpretation: Imports the modules needed to call the API, pause between readings, maintain the sliding window and calculate the statistics.

Step 2: Create a Function to Fetch the Bitcoin Price

```
def get_bitcoin_price():

    url = "https://api.coingecko.com/api/v3/simple/price"

    parameters = {
        "ids": "bitcoin",
```

```

        "vs_currencies": "usd"
    }

    try:
        response = requests.get(url, params=parameters, timeout=10)
        response.raise_for_status()
        data = response.json()

        if "bitcoin" in data and "usd" in data["bitcoin"]:
            return data["bitcoin"]["usd"]
        else:
            print("Unexpected API response:", data)
            return None

    except requests.exceptions.RequestException as e:
        print("Network / API Error:", e)
        return None

```

Interpretation: Calls the CoinGecko endpoint with the parameters `ids = bitcoin` and `vs_currencies = usd`. `raise_for_status()` reports HTTP errors, the response is validated before the price is read, and any network error is caught so that the program does not stop.

Step 3: Create the Sliding Window

```

window_size = 5
price_window = deque(maxlen=window_size)

```

Interpretation: Creates a deque that holds only the five most recent prices. When a sixth price is appended, the oldest one is removed automatically.

Step 4: Display the Start Message

```

print("Monitoring Bitcoin price...\n")

```

Interpretation: Indicates that monitoring has started.

Step 5: Fetch the Bitcoin Price 10 Times

```

for i in range(10):

    print(f"Reading {i + 1}")
    price = get_bitcoin_price()

    if price is not None:
        price_window.append(price)
        print("Current Price (USD):", price)
    else:
        print("Price could not be fetched")

    print("-" * 40)
    time.sleep(10)

```

Interpretation: Fetches the price ten times with a gap of ten seconds between readings, adding every valid price to the sliding window.

Step 6: Perform Statistical Analysis

```

if len(price_window) == window_size:
    print("Sliding Window :", list(price_window))

```

```

print("Mean      :", round(statistics.mean(price_window), 2))
print("Median   :", statistics.median(price_window))
print("Minimum  :", min(price_window))
print("Maximum  :", max(price_window))
print("SD       :", round(statistics.stdev(price_window), 2))

```

Interpretation: Statistics are calculated only after the window is completely filled with five prices. The standard deviation shows how much the price fluctuated within the recent window.

Complete Program

```

import requests
import time
import statistics
from collections import deque

def get_bitcoin_price():

    url = "https://api.coingecko.com/api/v3/simple/price"

    parameters = {
        "ids": "bitcoin",
        "vs_currencies": "usd"
    }

    try:
        response = requests.get(url, params=parameters, timeout=10)
        response.raise_for_status()
        data = response.json()

        if "bitcoin" in data and "usd" in data["bitcoin"]:
            return data["bitcoin"]["usd"]
        else:
            print("Unexpected API response:", data)
            return None

    except requests.exceptions.RequestException as e:
        print("Network / API Error:", e)
        return None

window_size = 5
price_window = deque(maxlen=window_size)

print("Monitoring Bitcoin price...\n")

for i in range(10):

    print(f"Reading {i + 1}")
    price = get_bitcoin_price()

    if price is not None:

        price_window.append(price)
        print("Current Price (USD):", price)
        if len(price_window) == window_size:

```

```

        print("Sliding Window :", list(price_window))
        print("Mean      :", round(statistics.mean(price_window), 2))
        print("Median    :", statistics.median(price_window))
        print("Minimum   :", min(price_window))
        print("Maximum   :", max(price_window))
        print("SD        :", round(statistics.stdev(price_window), 2))

    else:
        print("Price could not be fetched")

    print("-" * 40)
    time.sleep(10)

```

Sample Input

No input file is required. The data is fetched live from the API:

```

Endpoint   : https://api.coingecko.com/api/v3/simple/price
Params     : ids = bitcoin, vs_currencies = usd
Readings   : 10
Interval   : 10 seconds
Window     : 5

```

Expected Output

The values below are only a sample layout. The actual prices depend on the live market at the time the program is run.

Monitoring Bitcoin price...

```

Reading 1
Current Price (USD): 61240.0
-----
Reading 2
Current Price (USD): 61255.0
-----
Reading 3
Current Price (USD): 61198.0
-----
Reading 4
Current Price (USD): 61210.0
-----
Reading 5
Current Price (USD): 61262.0
Sliding Window : [61240.0, 61255.0, 61198.0, 61210.0, 61262.0]
Mean      : 61233.0
Median    : 61240.0
Minimum   : 61198.0
Maximum   : 61262.0
SD        : 27.96
-----
...
Reading 10

```

Interpretation

- The CoinGecko API returns the live Bitcoin price in USD.
- The price is fetched 10 times with a gap of 10 seconds between readings.

- A deque with maxlen = 5 keeps only the five most recent prices.
- When a sixth price arrives, the oldest price is dropped automatically.
- Statistics are calculated only after the window is completely filled.
- The standard deviation shows how much the price fluctuated in the recent window.
- The try-except block prevents the program from stopping when a network error occurs.

Result

The program successfully fetched real-time Bitcoin prices using the CoinGecko API. A sliding window of the latest five prices was maintained and updated continuously. Statistical measures such as mean, median, minimum, maximum and standard deviation were calculated for each window. The results displayed the current Bitcoin price along with updated statistics after every new price fetched, enabling continuous monitoring of market trends.